2026.10.15

§7. Sylow theorems.

如非交换单群中无正规子群，拆不了

Recall: if G finite abelian, $\#G = p_1^{r_1} \cdots p_m^{r_m}$, then $G = G_1 \times G_2 \times \cdots \times G_m$ ($\text{order } p_i^{r_i}$) 若是非交换有限群，则不能期待能拆成 p 群的直积或半直积.

退一步找 p 子群 → Sylow Thm.

DEF Let p be a prime number.

1) A <u>p-group</u> is a finite group G with $\#G = p^r$ for some r.

2) If G is a finite group with $\#G = p^r m$, $p \nmid m$.

a subgroup of order exactly p^r is called a <u>Sylow p-subgroup</u>.

Write $\mathrm{Syl}_p(G) := \{\text{Sylow } p\text{-subgps of } G\}$ $n_p := \#\mathrm{Syl}_p(G)$

THM (Sylow theorem) Let G be a finite group with $\#G = p^r m$, $p \nmid m$.

<u>First Sylow</u> Sylow p-subgp exist. 事实上，对 $0 \le k \le r$, G 的 p^k 阶子群均存在.

<u>Second Sylow</u> If P is a Sylow p-subgroup, $Q < G$ is a subgp of p-power order,

then $\exists g$, st. $Q < gPg^{-1}$ → also a Sylow p-subgp

(equivalently, all Sylow p-subgps are conjugate.

every subgp of p-power order is contained in a Sylow p-subgp.)

<u>Third Sylow</u> $n_p \equiv 1 \pmod p$ $n_p \mid m$.

Pf of first Sylow: induction on $\#G$. $\#G = 1$ ✓

为归纳，希望缩小群阶. 一种自然的方式是作商

找非平凡正规子群

中心.

Now assume first Sylow for all groups of order $< \#G = p^r m$.

<u>Case 1</u> If $p \mid \#Z(G)$ — finite abelian group.

then $Z(G) \simeq \underbrace{\mathbb{Z}/p^{r_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{r_s}\mathbb{Z}}_{p\text{-part } Z(G)_p} \times \cdots$

$\#Z(G)_p = p^{r'}$

Consider $\pi: G \to \overline{G} := G/Z(G)_p$ → has order $\frac{\#G}{p^{r'}} = p^{r-r'}m$

induction hypothesis $\Rightarrow \overline{G}$ contains a subgp $\overline{H}$ of subgp of order $p^{r-r'}$

$\Rightarrow H := \pi^{-1}(\overline{H}) < G$ is a subgp of order $p^{r-r'}p^{r'} = p^r \Rightarrow H$ ✓

Case 2 If $p \nmid \#Z(G)$ may assume $p \mid \#G$ (otherwise §13 ✓).

Class eq. $\#G = \#Z(G) + \sum [G : C_G(g_i)]$

($\#G$: div by p; $\#Z(G)$: not div by p)

$\Rightarrow \exists\, i$ s.t. $[G : C_G(g_i)]$ is not divisible by p and not 1.

$\Rightarrow C_G(g_i)$ has order $p^r m'$ for some $m' < m$.

induction hypothesis $\Rightarrow C_G(g_i)$ has a Sylow p-subgp H of order $p^r \Rightarrow H$ ✓ □

Pf of second Sylow. Let $P < G$ be a Sylow p-subgp. $Q < G$ be a subgp of p-power order.

if $\#Q = 1$ then $Q = \{1\} < P$. Now assume $\#Q = p^{r'}$ for some $r' > 0$. 欲证 $Q < gPg^{-1}$, 只要 $gg \in gP$, 只要 $ggP = gP$

Consider the action $Q \curvearrowright G/P = \{gP \mid g \in G\}$ by left translation. 即 gP 为左乘 g 的不动点. 自然考虑 $Q \curvearrowright G/P$

Then $\#(G/P) = \sum \#\text{orbits} = \sum \#(Q/\text{Stab}_Q(gP))$

($\#(G/P)$: not div by p; $\#(Q/\text{Stab}_Q(gP))$: p-power)

$\Rightarrow \exists$ an orbit of one element. Say of gP.

ie. $\forall g \in Q$, $ggP = gP \Rightarrow gg \in gP \Rightarrow g \in gPg^{-1} \Rightarrow Q < gPg^{-1}$.

COR All Sylow p-subgps are conjugate.

COR There is only one Sylow p-subgp $\Leftrightarrow$ A Sylow p-subgp is normal.

Recall $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$ ($H \triangle N_G(H)$)

COR If P is a Sylow p-subgp, then $N_G(N_G(P)) = N_G(P)$ and $N_G(P)$ contains a unique Sylow p-subgp (namely P).

Pf $P \triangleleft N_G(P) \Rightarrow P$ is the unique Sylow p-subgp of $N_G(P)$.

$\forall n \in N_G(N_G(P))$, $n N_G(P) n^{-1} = N_G(P)$ (nPn^{-1}, P) $\Rightarrow nPn^{-1} = P \Rightarrow n \in N_G(P)$ □

Pf of third Sylow. $n_p = \#$ of Sylow subgps.

Since all Sylow p-subgps are conjugate, $G \overset{Ad}{\curvearrowright} \{\text{Sylow } p\text{-subgps}\}$ transitive action

$\Rightarrow n_p = \#(G/\text{Stab}_G(P)) = [G : N_G(P)] = \dfrac{p^r \cdot m}{p^r \cdot m'}$ ($\text{Stab}_G(P) = N_G(P)$; $P \triangle N_G(P)$) $m' = [N_G(P) : P] \Rightarrow n_p \mid m$. $n_p = [G : N_G(P)]$

Consider $P \overset{Ad}{\curvearrowright} \{\text{Sylow } p\text{-subgps}\} \Rightarrow n_p = \sum_i \#(P/\text{Stab}_P(P_i))$

if $\text{Stab}_P(P_i) \neq P$, $\#(P/\text{Stab}_P(P_i))$ is a nontrivial p-power.

if $Stab_P(P_i)=P$, $P<N_G(P_i)$. But $N_G(P_i)$ has a unique Sylow p-subgp (namely P_i)

$\Rightarrow P=P_i \Rightarrow n_p\equiv 1 \pmod p$. □

Applications

1). Groups of order pq p,q prime p<q.

Q := a Sylow q-subgp $\simeq \mathbb{Z}/q\mathbb{Z}$. $n_q\equiv 1 \pmod q$ $n_q\mid p$ $p<q \Rightarrow n_q=1 \Rightarrow Q\lhd G$.

P := a Sylow p-subgp $\simeq \mathbb{Z}/p\mathbb{Z}$. $P\cap Q=\{1\}$ $PQ=G$

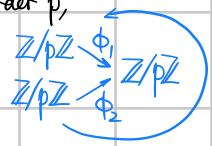
$\Rightarrow G\simeq Q\rtimes P\simeq \mathbb{Z}/q\mathbb{Z}\rtimes_\phi \mathbb{Z}/p\mathbb{Z}$.

Want $\phi: \mathbb{Z}/p\mathbb{Z} \xrightarrow{\text{hom.}} Aut(\mathbb{Z}/q\mathbb{Z})\simeq(\mathbb{Z}/q\mathbb{Z})^\times$ cycle of order q-1.

if $p\nmid q-1$ then ϕ is necessarily trivial $\Rightarrow G\simeq \mathbb{Z}/q\mathbb{Z}\times\mathbb{Z}/p\mathbb{Z}\cong\mathbb{Z}/pq\mathbb{Z}$.

n阶循环群C_n有且仅有1个d阶子群，其中$d\mid n$.

if $p\mid q-1$ then $\exists$ nontrivial ϕ since $(\mathbb{Z}/q\mathbb{Z})^\times$ contains a unique cyclic subgp of order p,



$\operatorname{im}\phi_1=\operatorname{im}\phi_2=\mathbb{Z}/p\mathbb{Z}$

考虑 $\phi_1^{-1}\circ\phi_2$.

exercise: get isomorphism G for different nontrivial ϕ.

$\Rightarrow$ either $G\simeq\mathbb{Z}/pq\mathbb{Z}$ or $G\simeq\mathbb{Z}/q\mathbb{Z}\rtimes_\phi(\text{nontrivial})\ \mathbb{Z}/p\mathbb{Z}$.

2). No simple groups of order $132=3\times4\times11$.

Pf. Suppose G is simple $\Rightarrow$ no nontrivial normal subgp.

否则它是它真子群.

$n_{11}\equiv 1 \pmod{11}$ $n_{11}\mid 12$ $\boxed{n_{11}\neq 1}$ $\Rightarrow n_{11}=12$. $\Rightarrow$ at least 12×10 elements of order 11.

$n_3\equiv 1 \pmod 3$ $n_3\mid 44$ $n_3\neq 1 \Rightarrow n_3=4$ or 22. $\Rightarrow$ at least 4×2 elements of order 3.

$n_2\neq 1 \Rightarrow \exists$ at least two Sylow 2-subgps P_1, P_2. $\Rightarrow$ at least 6 elements whose order divides 4.

$120+8+6>132$ Too many elements! □

3). Let G be a finite group. $N\lhd G$ $\pi: G\to G/N$. $\#G=p^r m$, $\#N=p^{r'}m'$, $\#(G/N)=p^{r-r'}\frac{m}{m'}$.

Let H be a Sylow p-subgp of G, $\#H=p^r$.

Then $\pi(H)$ is a Sylow p-subgp of G/N, $\ker(\pi|_H: H\to\pi(H))=H\cap N$ is a Sylow p-subgp of N.

Pf. First isom thm $\Rightarrow H/H\cap N\simeq\pi(H)$. $\#H=p^r=\#(H\cap N)\cdot\#\pi(H)$.

But $\#(H\cap N)$ divides $\#N=p^{r'}m'$, $\#\pi(H)$ divides $\#(G/N)=p^{r-r'}\frac{m}{m'}$.

$\Rightarrow$ Necessarily $\#(H\cap N)=p^{r'}$, $\#\pi(H)=p^{r-r'}$. $\square$

4). If G of order 105 contains a normal Sylow 3-subgp P_3,

then $G\simeq \mathbb{Z}/105\mathbb{Z}$.

Pf. $n_5\equiv 1 \ (\mathrm{mod}\ 5)$ $n_5\mid 3\times 7$ $\Rightarrow n_5=1$ or 21.

$n_7\equiv 1 \ (\mathrm{mod}\ 7)$ $n_7\mid 3\times 5$ $\Rightarrow n_7=1$ or 15.

Now consider $G \overset{Ad}{\curvearrowright} P_3$ (P_3 normal) $\quad Ad: G\to Aut(P_3)\simeq Aut(\mathbb{Z}/3\mathbb{Z})\simeq\{\pm 1\}$

But $\#G=105$ is odd $\Rightarrow Im(Ad)=\{1\}$. $\Rightarrow$ every element of G commutes with every element of P_3.

$\Rightarrow P_3<Z(G)$. Recall the proof of third Sylow.

$n_5=\#G/\#N_G(P_5)$ ($N_G(P_5)$ contains P_5 and P_3.) $\Rightarrow n_5\neq 21$ $n_5=1$

$n_7=\#G/\#N_G(P_7)$ ($N_G(P_7)$ contains P_7 and P_3.) $\Rightarrow n_7=1$.

$\Rightarrow P_3, P_5, P_7$ are all normal $\Rightarrow G\simeq \mathbb{Z}/3\mathbb{Z}\times\mathbb{Z}/5\mathbb{Z}\times\mathbb{Z}/7\mathbb{Z}\simeq\mathbb{Z}/105\mathbb{Z}$. $\square$

§8. Commutator subgroup, nilpotent groups, solvable groups, p-groups.

DEF For $x,y\in G$ define $[x,y]=x^{-1}y^{-1}xy$, called the commutator (交换子/换位子) of x,y. 刻画群有多"不交换".

(Some people use $xyx^{-1}y^{-1}$)

(in particular, $xy=yx[x,y]$. $xy=yx\Leftrightarrow[x,y]=1$. $g[x,y]g^{-1}=[gxg^{-1},gyg^{-1}]$.)

$G'=G^{(1)}=G^{der}=[G,G]:=\langle [x,y]\mid x,y\in G\rangle$ (subgp generated)

called the commutator/derived subgroup (交换子群/导子群) of G

! Not every element of G' is a commutator.

Then $G'\lhd G$ normal subgp and G/G' is abelian.

($xG'\cdot yG'\overset{?}{=}yG'\cdot xG' \Leftrightarrow x^{-1}y^{-1}xy\in G'$)

$xG'\cdot yG'=xyG'$ $\quad yG'\cdot xG'=yxG'$